In [61]:

```
import pandas as pd
df=pd.read_csv('CovidData.csv',encoding='latin-1')
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 438960 entries, 0 to 438959
Data columns (total 8 columns):
 #   Column             Non-Null Count   Dtype
---  ------             --------------   -----
 0   ï»¿Date_reported   438960 non-null  object
 1   Country_code       437131 non-null  object
 2   Country            438960 non-null  object
 3   WHO_region         438960 non-null  object
 4   New_cases          207884 non-null  float64
 5   Cumulative_cases   438960 non-null  int64
 6   New_deaths         208824 non-null  float64
 7   Cumulative_deaths  438960 non-null  int64
dtypes: float64(2), int64(2), object(4)
memory usage: 26.8+ MB
None
```

In [62]:

```
print(df.head())
```

```
  ï»¿Date_reported Country_code               Country WHO_region  New_cases  \
0       2020-01-04           SB       Solomon Islands        WPR        0.0
1       2020-01-04           LK             Sri Lanka       SEAR        0.0
2       2020-01-04           SY  Syrian Arab Republic        EMR        NaN
3       2020-01-04           TJ            Tajikistan        EUR        NaN
4       2020-01-04           TH              Thailand       SEAR        0.0

   Cumulative_cases  New_deaths  Cumulative_deaths
0                 0         0.0                  0
1                 0         0.0                  0
2                 0         NaN                  0
3                 0         NaN                  0
4                 0         0.0                  0
```

In [63]:
```
print(df.describe().round())
```

```
       New_cases  Cumulative_cases  New_deaths  Cumulative_deaths
count   207884.0          438960.0    208824.0           438960.0
mean      3739.0         1907201.0        34.0            20771.0
std      48848.0         8136020.0       209.0            84119.0
min     -65079.0               0.0     -3520.0                0.0
25%          3.0            5393.0         0.0               34.0
50%         62.0           50948.0         0.0              650.0
75%        690.0          600138.0         7.0             7475.0
max    6966046.0       103436829.0     44047.0          1212060.0
```

In [64]:
```
print(df.isnull().sum())
```

```
ï»¿Date_reported          0
Country_code           1829
Country                   0
WHO_region                0
New_cases            231076
Cumulative_cases          0
New_deaths           230136
Cumulative_deaths         0
dtype: int64
```

In [65]:
```
print(df.isnull().sum())
```

```
ï»¿Date_reported          0
Country_code           1829
Country                   0
WHO_region                0
New_cases            231076
Cumulative_cases          0
New_deaths           230136
Cumulative_deaths         0
dtype: int64
```

In [66]:
```
df.columns.tolist()
```

Out[66]:
```
['ï»¿Date_reported',
 'Country_code',
 'Country',
 'WHO_region',
 'New_cases',
 'Cumulative_cases',
 'New_deaths',
 'Cumulative_deaths']
```

In [67]:
```
df.dtypes
```

Out[67]:
```
ï»¿Date_reported      object
Country_code          object
Country               object
WHO_region            object
New_cases            float64
Cumulative_cases       int64
New_deaths           float64
Cumulative_deaths      int64
dtype: object
```

In [68]:
```
df.columns=df.columns.str.strip().str.lower()
print(df.columns.tolist())
```

```
['ï»¿date_reported', 'country_code', 'country', 'who_region', 'new_cases', 'cumulative_cases', 'new_deaths', 'cumulative_deaths']
```

In [69]:
```
df=df.rename(columns={'ï»¿date_reported':'date_reported'})
```

In [70]:
```
df['date_reported']=pd.to_datetime(df['date_reported'])
print(df.dtypes)
```

```
date_reported        datetime64[ns]
country_code                 object
country                      object
who_region                   object
new_cases                   float64
cumulative_cases              int64
new_deaths                  float64
cumulative_deaths             int64
dtype: object
```

In [71]:
```
print(df.duplicated().sum())
```

```
0
```

In [72]:
```
df.duplicated(subset='country_code').sum()
```

Out[72]:
```
438720
```

In [73]:
```
df.duplicated(subset=['country_code','date_reported']).sum()
```

Out[73]:
```
0
```

In [74]:
```
df['country_code'].value_counts()
```

Out[74]:
```
SB    1829
CD    1829
PR    1829
MD    1829
MF    1829
      ...
MN    1829
MS    1829
NO    1829
BL    1829
YE    1829
Name: country_code, Length: 239, dtype: int64
```

In [75]:
```
df['country_code'].nunique()
```

Out[75]:
```
239
```

In [76]:
```
df.isnull().sum()
```

Out[76]:
```
date_reported             0
country_code           1829
country                   0
who_region                0
new_cases            231076
cumulative_cases          0
new_deaths           230136
cumulative_deaths         0
dtype: int64
```

In [77]:
```
df['country_code']=df['country_code'].fillna('UNKNOWN')
```

In [90]:
```
(df['country_code']=="UNKNOWN").sum()
```

Out[90]:
```
1829
```

In [91]:
```
df=df.fillna(0)
print(df.isnull().sum())
```

```
date_reported        0
country_code         0
country              0
who_region           0
new_cases            0
cumulative_cases     0
new_deaths           0
cumulative_deaths    0
dtype: int64
```

In [92]:
```
print(df.dtypes)
```

```
date_reported        datetime64[ns]
country_code                 object
country                      object
who_region                   object
new_cases                   float64
cumulative_cases              int64
new_deaths                  float64
cumulative_deaths             int64
dtype: object
```

In [93]:
```
df=df.astype({'new_cases':int,'new_deaths':int})
print(df.dtypes)
```

```
date_reported        datetime64[ns]
country_code                 object
country                      object
who_region                   object
new_cases                     int32
cumulative_cases              int64
new_deaths                    int32
cumulative_deaths             int64
dtype: object
```

In [94]:
```
df.head()
```

Out[94]:

	date_reported	country_code	country	who_region	new_cases	cumulative_cases	new_deaths	cumulative_deaths
0	2020-01-04	SB	Solomon Islands	WPR	0	0	0	0
1	2020-01-04	LK	Sri Lanka	SEAR	0	0	0	0
2	2020-01-04	SY	Syrian Arab Republic	EMR	0	0	0	0
3	2020-01-04	TJ	Tajikistan	EUR	0	0	0	0
4	2020-01-04	TH	Thailand	SEAR	0	0	0	0

In [95]:
```
df.describe().round()
```

Out[95]:

	new_cases	cumulative_cases	new_deaths	cumulative_deaths
count	438960.0	438960.0	438960.0	438960.0
mean	1771.0	1907201.0	16.0	20771.0
std	33667.0	8136020.0	145.0	84119.0
min	-65079.0	0.0	-3520.0	0.0
25%	0.0	5393.0	0.0	34.0
50%	0.0	50948.0	0.0	650.0
75%	47.0	600138.0	0.0	7475.0
max	6966046.0	103436829.0	44047.0	1212060.0

In [96]:
```
(df['new_cases']<0).sum()
```

Out[96]: 71

In [97]:
```
df.head(2)
```

Out[97]:

	date_reported	country_code	country	who_region	new_cases	cumulative_cases	new_deaths	cumulative_deaths
0	2020-01-04	SB	Solomon Islands	WPR	0	0	0	0
1	2020-01-04	LK	Sri Lanka	SEAR	0	0	0	0

In [98]:
```
df
```

Out[98]:

	date_reported	country_code	country	who_region	new_cases	cumulative_cases	new_deaths	cumulative_deaths
0	2020-01-04	SB	Solomon Islands	WPR	0	0	0	0
1	2020-01-04	LK	Sri Lanka	SEAR	0	0	0	0
2	2020-01-04	SY	Syrian Arab Republic	EMR	0	0	0	0
3	2020-01-04	TJ	Tajikistan	EUR	0	0	0	0
4	2020-01-04	TH	Thailand	SEAR	0	0	0	0
...	...	...	...	...	...	...	...	...
438955	2025-01-05	TT	Trinidad and Tobago	AMR	0	191496	0	4390
438956	2025-01-05	TN	Tunisia	EMR	0	1153361	0	29423
438957	2025-01-05	UA	Ukraine	EUR	0	5541740	0	109925
438958	2025-01-05	UY	Uruguay	AMR	0	1042557	0	7694
438959	2025-01-05	YE	Yemen	EMR	0	11945	0	2159

438960 rows × 8 columns

In [99]:
```
df['new_cases'].nunique()
```

Out[99]:
```
18334
```

In [100]:
```
df['country'].value_counts()
```

Out[100]:
```
Solomon Islands        1829
Sri Lanka              1829
Puerto Rico            1829
Republic of Moldova    1829
Saint Martin           1829
                       ...
Mongolia               1829
Montserrat             1829
Norway                 1829
Saint Barthelemy       1829
Yemen                  1829
Name: country, Length: 240, dtype: int64
```

In [101]:
```
df.sample(10)
```

Out[101]:

	date_reported	country_code	country	who_region	new_cases	cumulative_cases	new_deaths	cumulative_deaths
355521	2024-01-24	RE	RÃ©union	AFR	0	494595	0	921
136886	2021-07-27	NO	Norway	EUR	315	136464	0	803
225552	2022-07-31	JE	Jersey, Channel Islands	EUR	24	58121	0	133
437501	2024-12-30	SV	El Salvador	AMR	0	201967	0	4230
242513	2022-10-10	MT	Malta	EUR	0	114867	0	827
106470	2021-03-22	VU	Vanuatu	WPR	0	3	0	0
149894	2021-09-19	GL	Greenland	EUR	10	464	0	0
6780	2020-02-01	RU	Russian Federation	EUR	2	2	0	0
281831	2023-03-23	AW	Aruba	AMR	0	44099	0	287
208204	2022-05-20	ST	Sao Tome and Principe	AFR	0	5970	0	73

In [102]:
```
df.to_csv("Cleaned_CovidData.csv")
```